# Discrete Mathematics

CSE 4203

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## Chapter 4 Number Theory and Cryptography

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# CHAPTER 4: Number Theory and Cryptography

# Division of Integers

## Definition 1: Divisibility

Let $a$ and $b$ be integers with $a \neq 0$.
We say that $a$ **divides** $b$ (written as $a \mid b$) if there exists an integer $c$ such that:

$$b = ac$$

Equivalently, $\frac{b}{a}$ is an integer.

- If $a \mid b$, then:
  - $a$ is a **divisor** or **factor** of $b$
  - $b$ is a **multiple** of $a$
- If $a \nmid b$, then $\frac{b}{a} \notin \mathbb{Z}$

**Using quantifiers:**

$$a \mid b \Leftrightarrow \exists c \in \mathbb{Z},\ ac = b$$

# Example: Divisibility

**Example 1:** Determine whether $3 \mid 7$ and $3 \mid 12$.

- $\frac{7}{3} = 2.\overline{3} \notin \mathbb{Z} \Rightarrow 3 \nmid 7$
- $\frac{12}{3} = 4 \in \mathbb{Z} \Rightarrow 3 \mid 12$

# Theorem 1: Properties of Divisibility

**Let** $a, b, c$ **be integers, where** $a \neq 0$. **Then:**

**(i)** If $a \mid b$ and $a \mid c$, then $a \mid (b + c)$.

**(ii)** If $a \mid b$, then $a \mid bc$ for all integers $c$.

**(iii)** If $a \mid b$ and $b \mid c$, then $a \mid c$.

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**(ii)** If $a \mid b$, then $a \mid bc$ for all integers $c$.

**(iii)** If $a \mid b$ and $b \mid c$, then $a \mid c$.

**Interpretation**

These rules help simplify proofs involving divisibility in number theory.

# Theorem 1 (i): Proof

**Statement:**
Let $a, b, c \in \mathbb{Z}$, with $a \neq 0$.
**If** $a \mid b$ and $a \mid c$, **then** $a \mid (b + c)$.

# Theorem 1 (i): Proof

**Statement:**
Let $a, b, c \in \mathbb{Z}$, with $a \neq 0$.
**If** $a \mid b$ and $a \mid c$, **then** $a \mid (b + c)$.
**Proof:**

- Since $a \mid b$, there exists an integer $m$ such that:

$$b = am$$

- Since $a \mid c$, there exists an integer $n$ such that:

$$c = an$$

- Then:

$$b + c = am + an = a(m + n)$$

- Since $m + n \in \mathbb{Z}$, we have:

$$a \mid (b + c) \quad \blacksquare$$

# Example of Theorem 1 (i)

**Let** $a = 5$**,** $b = 20$**,** $c = 35$**.**

- $5 \mid 20$ since $20 = 5 \times 4$
- $5 \mid 35$ since $35 = 5 \times 7$
- So, $5 \mid (20 + 35) = 5 \mid 55$
- And $55 = 5 \times 11$, so $\boxed{5 \mid 55}$

# Example of Theorem 1 (i)

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- So, $5 \mid (20 + 35) = 5 \mid 55$
- And $55 = 5 \times 11$, so $\boxed{5 \mid 55}$

**Conclusion:** The example confirms that if $a \mid b$ and $a \mid c$, then $a \mid (b + c)$

# 4.1.3 The Division Algorithm

**Theorem 2: Division Algorithm**
Let $a$ be any integer and $d$ a positive integer.
**Then there exist unique integers** $q$ and $r$ such that:

$$a = dq + r \quad \text{where } 0 \leq r < d$$

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**Definition 2: Terminology**

- $a$: Dividend
- $d$: Divisor
- $q$: Quotient
- $r$: Remainder

**Notation:**

$$q = a \div d, \quad r = a \bmod d$$

# Example: Division Algorithm

**Example 3:** What are the quotient and remainder when 101 is divided by 11?

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$$101 = 11 \cdot 9 + 2$$

- Quotient: $q = 101 \div 11 = 9$
- Remainder: $r = 101 \bmod 11 = 2$

**Check:** $dq + r = 11 \cdot 9 + 2 = 101 \quad \checkmark$

# Example: Division Algorithm

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**Verification:** $3q + r = 3(-4) + 1 = -12 + 1 = -11 \quad \checkmark$

# 4.1.4 Congruence Modulo $n$

**Definition:**

Let $a, b \in \mathbb{Z}$ and $n \in \mathbb{Z}^+$.
We say that $a$ is **congruent to** $b$ modulo $n$, written as:

$$a \equiv b \pmod{n}$$

if and only if $n \mid (a - b)$

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**Example 1:**

$$38 \equiv 14 \pmod{12} \quad \text{since } 38 - 14 = 24 \text{ and } 12 \mid 24$$

**Theorem 4**

Let $m$ be a positive integer. The integers $a$ and $b$ are congruent modulo $n$ if and only if there exists an integer $k$ such that

$$a = b + kn.$$

# Theorem: Properties of Congruence

Let $a, b, c \in \mathbb{Z}$ and $n \in \mathbb{Z}^+$. Then:

- **Reflexive:** $a \equiv a \pmod{n}$
- **Symmetric:** If $a \equiv b \pmod{n}$, then $b \equiv a \pmod{n}$
- **Transitive:** If $a \equiv b \pmod{n}$ and $b \equiv c \pmod{n}$, then $a \equiv c \pmod{n}$

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**Why?** All follow from divisibility:

$$a \equiv b \pmod{n} \Leftrightarrow n \mid (a - b)$$

# Theorem: Properties of Congruence

Let $a \equiv b \pmod{n}$ and $c \equiv d \pmod{n}$. Then:

1. **Addition:** $a + c \equiv b + d \pmod{n}$
2. **Subtraction:** $a - c \equiv b - d \pmod{n}$
3. **Multiplication:** $ac \equiv bd \pmod{n}$

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**Proof:** Let $r_1 = a \bmod n$ and $r_2 = b \bmod n$. Then there exist integers $k_1$ and $k_2$ such that:

$$a = r_1 + k_1 n \quad \text{and} \quad b = r_2 + k_2 n$$

$$\Rightarrow a + b = (r_1 + r_2) + (k_1 + k_2)n \Rightarrow (a + b) \bmod n = (r_1 + r_2) \bmod n$$

$$\text{Similarly,} \quad ab = (r_1 + k_1 n)(r_2 + k_2 n)$$

$$= r_1 r_2 + n(\cdots) \Rightarrow ab \bmod n = (r_1 r_2) \bmod n$$

# Equivalence Classes Modulo $n$

**Definition:**
Let $a \in \mathbb{Z}$ and $n \in \mathbb{Z}^+$. The set of all integers congruent to $a$ modulo $n$ is called the **equivalence class of** $a$ modulo $n$, denoted by:

$$[a]_n = \{x \in \mathbb{Z} \mid x \equiv a \pmod{n}\}$$

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**Examples:**

- $[0]_5 = \{\ldots, -10, -5, 0, 5, 10, 15, \ldots\}$
- $[2]_5 = \{\ldots, -8, -3, 2, 7, 12, 17, \ldots\}$

# Equivalence Classes Modulo $n$

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**Examples:**

- $[0]_5 = \{\dots, -10, -5, 0, 5, 10, 15, \dots\}$
- $[2]_5 = \{\dots, -8, -3, 2, 7, 12, 17, \dots\}$

**Key Idea:** Every integer belongs to one of the $n$ equivalence classes:

$$[0]_n, [1]_n, \dots, [n-1]_n$$

# Theorem 3: Congruence and Remainders

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Let $a$ and $b$ be integers, and let $m$ be a positive integer. Then:

$$\boxed{a \equiv b \pmod{m} \iff a \bmod m = b \bmod m}$$

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**Proof:**

Suppose $a \equiv b \pmod{m} \Rightarrow m \mid (a - b)$

$\Rightarrow a = b + km$ for some $k \in \mathbb{Z}$

So $a$ and $b$ have the same remainder when divided by $m$

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Suppose $a \bmod m = b \bmod m = r$

Then $a = mq_1 + r$, $b = mq_2 + r$, for some $q_1, q_2 \in \mathbb{Z}$

So $a - b = m(q_1 - q_2) \Rightarrow m \mid (a-b)$

$\Rightarrow a \equiv b \pmod{m}$

$$\Rightarrow \boxed{a \equiv b \pmod{m} \iff a \bmod m = b \bmod m} \quad \blacksquare$$

# Modular Arithmetic: Corollary

**Corollary (Modular Addition and Multiplication)**

*Let $m$ be a positive integer and let $a$ and $b$ be integers. Then:*

$$(a+b) \bmod m = ((a \bmod m) + (b \bmod m)) \bmod m$$

*and*

$$(ab) \bmod m = ((a \bmod m)(b \bmod m)) \bmod m$$

# Example 7: Modular Arithmetic

## Example (Evaluating $(193 \bmod 31)^4 \bmod 23$)

To compute $(193 \bmod 31)^4 \bmod 23$, we proceed as follows:

- First, evaluate $193 \bmod 31$.
  Since $193 = 6859 = 221 \cdot 31 + 8$,

  $$193 \bmod 31 = 6859 \bmod 31 = 8$$

- So,

  $$(193 \bmod 31)^4 \bmod 23 = 8^4 \bmod 23$$

- Next, compute $8^4 = 4096$. Since $4096 = 178 \cdot 23 + 2$,

  $$4096 \bmod 23 = 2$$

- Therefore,

  $$(193 \bmod 31)^4 \bmod 23 = 2$$

# 4.1.5 Arithmetic Modulo $m$

**Definition:** Let $m$ be a positive integer. Arithmetic modulo $m$ involves performing arithmetic operations (addition, subtraction, multiplication) on integers and then reducing the result modulo $m$.

**Basic Rules:**

- $(a + b) \bmod m = ((a \bmod m) + (b \bmod m)) \bmod m$
- $(a - b) \bmod m = ((a \bmod m) - (b \bmod m)) \bmod m$
- $(a \cdot b) \bmod m = ((a \bmod m) \cdot (b \bmod m)) \bmod m$

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**Note:** Modular division is more complex and usually requires computing a modular inverse.

**Example:**

$$(17 + 26) \bmod 5 = 43 \bmod 5 = 3$$

$$(7 \cdot 9) \bmod 4 = 63 \bmod 4 = 3$$

# 4.1.5 Arithmetic Modulo $m$: Definitions and Example

**Definition: Arithmetic in $\mathbb{Z}_m$** Let $\mathbb{Z}_m = \{0, 1, 2, \ldots, m-1\}$. For $a, b \in \mathbb{Z}_m$, we define:

- **Addition modulo $m$:**

$$a +_m b = (a + b) \bmod m$$

- **Multiplication modulo $m$:**

$$a \cdot_m b = (a \cdot b) \bmod m$$

**Example 8:** Compute $7 +_{11} 9$ and $7 \cdot_{11} 9$:

$$7 +_{11} 9 = (7 + 9) \bmod 11 = 16 \bmod 11 = 5$$

$$7 \cdot_{11} 9 = (7 \cdot 9) \bmod 11 = 63 \bmod 11 = 8$$

**Answer:** $7 +_{11} 9 = 5, 7 \cdot_{11} 9 = 8$

# Properties of Arithmetic Modulo $m$ in $\mathbb{Z}_m$

**The operations $+_m$ and $\cdot_m$ satisfy:**

- **Closure:** If $a, b \in \mathbb{Z}_m$, then $a +_m b \in \mathbb{Z}_m$ and $a \cdot_m b \in \mathbb{Z}_m$.
- **Associativity:** $(a +_m b) +_m c = a +_m (b +_m c)$ $(a \cdot_m b) \cdot_m c = a \cdot_m (b \cdot_m c)$
- **Commutativity:** $a +_m b = b +_m a$,$a \cdot_m b = b \cdot_m a$
- **Identity Elements:** 0 is the additive identity: $a +_m 0 = a$ 1 is the multiplicative identity: $a \cdot_m 1 = a$
- **Additive Inverses:** For $a \neq 0$, the additive inverse is $m - a$: $a +_m (m - a) = 0$,$0 +_m 0 = 0$
- **Distributivity:** $a \cdot_m (b +_m c) = (a \cdot_m b) +_m (a \cdot_m c)$

# Theorem and Proof

**Theorem:** Let $a, b, c, m$ be integers such that $m \geq 2$, $c > 0$, and

$$a \equiv b \pmod{m}.$$

Then

$$ac \equiv bc \pmod{mc}.$$

**Proof:** From $a \equiv b \pmod{m}$, we know

$$m \mid (a - b),$$

which means there exists an integer $k$ such that

$$a - b = km.$$

Multiplying both sides by $c$, we get

$$c(a - b) = c \cdot km = k(mc).$$

Rearranging,

$$ac - bc = k(mc).$$

# Analysis of Functions $f(a) = a \div d$ and $g(a) = a \bmod d$

Let $d$ be a fixed positive integer. Consider:

$$f(a) = a \div d, \quad g(a) = a \bmod d,$$

where the domain and codomain are the set of all integers $\mathbb{Z}$.

1. **Function** $f(a) = a \div d$:
   - **One-to-one?**
     No. Different integers can have the same quotient.
     Example: For $d = 2$, $f(2) = 1$ and $f(3) = 1$.
   - **Onto?**
     Yes. For any integer $q$, choosing $a = q \times d$ yields $f(a) = q$.

2. **Function** $g(a) = a \bmod d$:
   - **One-to-one?**
     No. Many integers share the same remainder.
     Example: $g(2) = 2$ and $g(2 + d) = 2$.
   - **Onto?**

# Theorem: Powers in Modular Arithmetic

**Theorem:** Let $a, b, k, m$ be integers with $k \geq 1$ and $m \geq 2$. If

$$a \equiv b \pmod{m},$$

then

$$a^k \equiv b^k \pmod{m}.$$

**Proof (by induction on $k$):**
**Base case $k = 1$:**

$$a^1 = a \equiv b = b^1 \pmod{m}$$

True by assumption.

# Proof (continued)

**Inductive step:** Assume for some $n \geq 1$,

$$a^n \equiv b^n \pmod{m}.$$

We need to show:

$$a^{n+1} \equiv b^{n+1} \pmod{m}.$$

Using the inductive hypothesis and the initial congruence,

$$a^{n+1} = a^n \cdot a \equiv b^n \cdot b = b^{n+1} \pmod{m}.$$

**Conclusion:** By induction,

$$a^k \equiv b^k \pmod{m} \quad \text{for all } k \geq 1.$$

□